

Lab 9: Deep Neural Network for MNIST Classification

Aim

To implement a **Deep Neural Network (DNN)** for handwritten digit classification using the **MNIST dataset**.

Algorithm

1. Import the required libraries.
2. Load the MNIST dataset.
3. Normalize the pixel values.
4. Create a Deep Neural Network with hidden layers.
5. Train the model using the training dataset.
6. Evaluate the model using the test dataset.
7. Display the test accuracy.

Program

```
# Lab 9: Deep Neural Network for MNIST Classification
```

```
import tensorflow as tf

from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten

# Load MNIST dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0
```

```

# Create the Deep Neural Network
model = Sequential([
    Flatten(input_shape=(28, 28)),
    Dense(128, activation='relu'),
    Dense(64, activation='relu'),
    Dense(10, activation='softmax')
])

# Compile the model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Train the model
model.fit(
    X_train,
    y_train,
    epochs=5,
    batch_size=32
)

# Evaluate the model
loss, accuracy = model.evaluate(X_test, y_test)

print("Test Accuracy:", accuracy)

```

Sample Output

Test Accuracy: 0.97

The exact accuracy may vary slightly each time you run it.

Result

Thus, the **Deep Neural Network** was successfully implemented using the **MNIST dataset** and used to classify handwritten digits with high accuracy.